



RESEARCH METHODOLOGY

LECTURE 1



CSJM
UNIVERSITY
KANPUR



PHILOSOPHIZING RESEARCH

COURSE ORIENTATION PROGRAM

in

RESEARCH IN EDUCATION

M.A. | M.Ed. | Ph.D. | NET/JRF



Philosophy



Thinking



Research
Questions



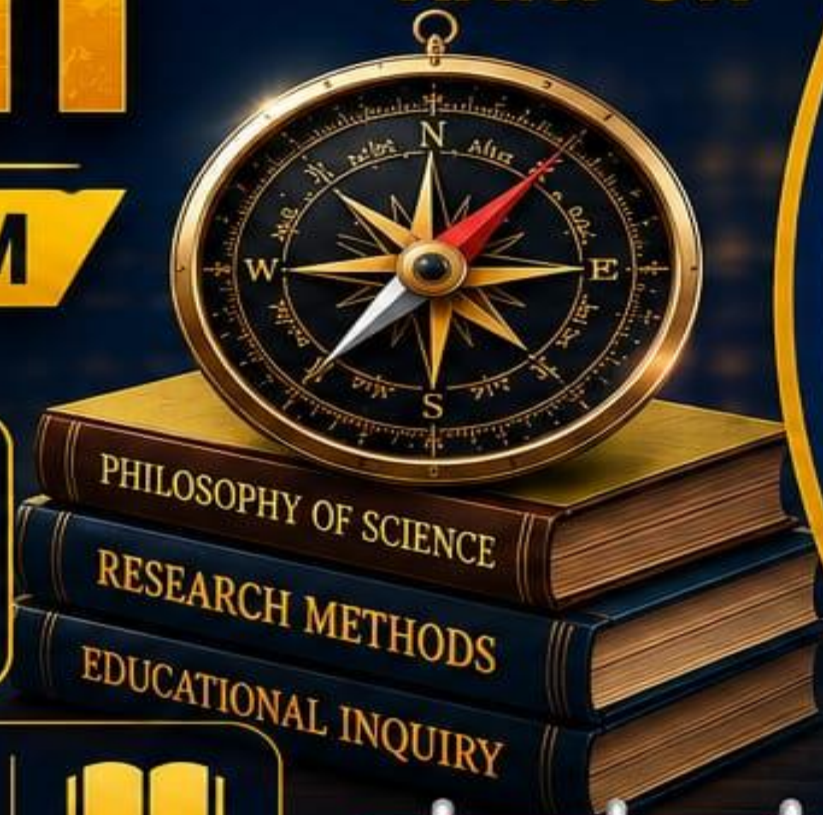
Methodology



Findings



Knowledge



DATE

10/07/2026



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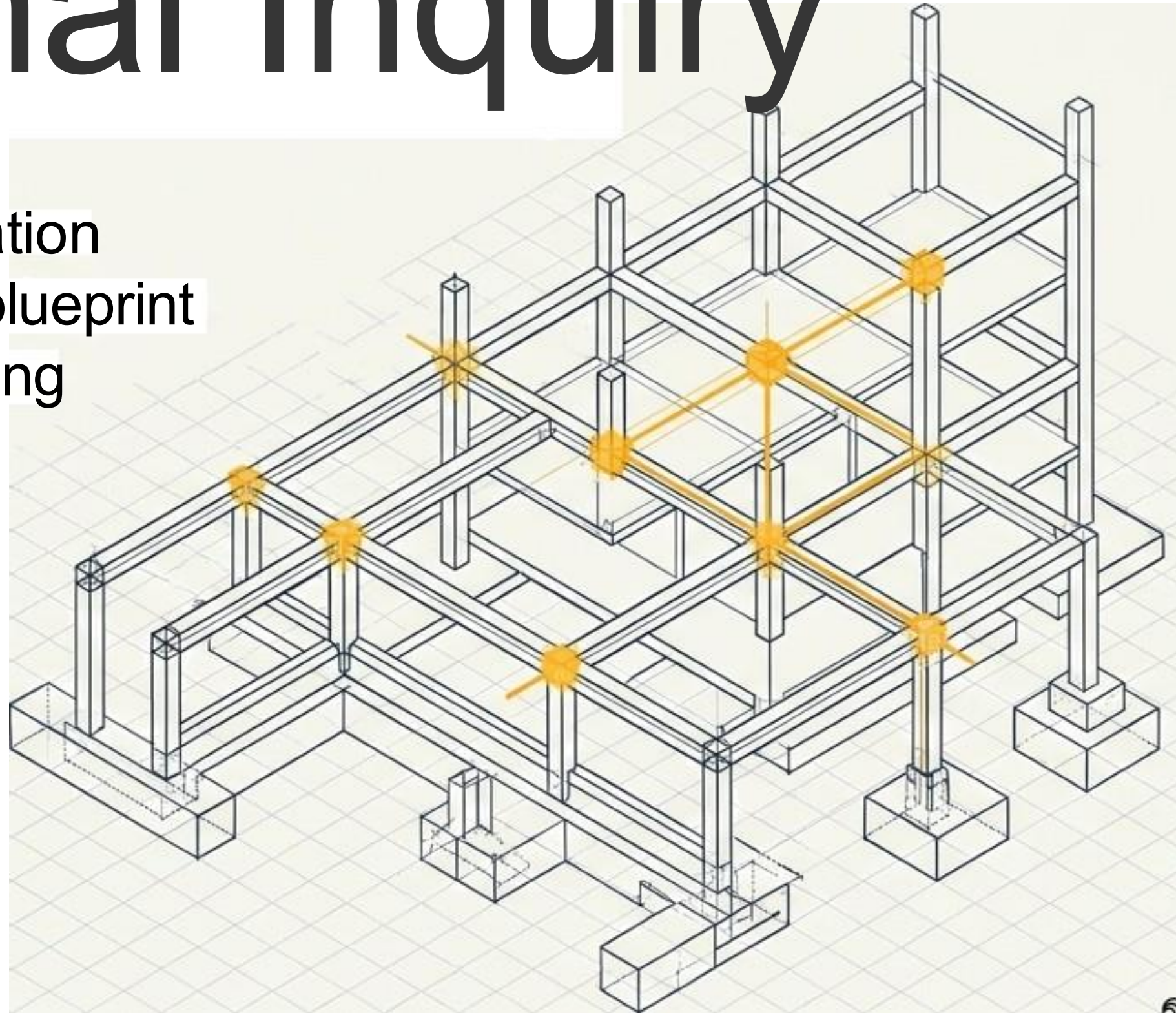
COMPLETE CONCEPT
WITH EXAMPLES



FOR Ph.D. COURSEWORK,
NET/JRF & FACULTY DEVELOPMENT

Constructing Educational Inquiry

MED104: Research Methods in Education (General Perspectives). A systematic blueprint for designing, structuring, and executing educational research.



The Master Plan for the Prospective Teacher Educator

Educational research is not a passive observation; it is a constructed inquiry. This course equips you to transition from consumer of educational theory to architect of new knowledge.



1. Grasping the Scientific Method & Paradigms (Surveying the land)



2. Distinguishing Quantitative & qualitative Approaches (Choosing the materials)



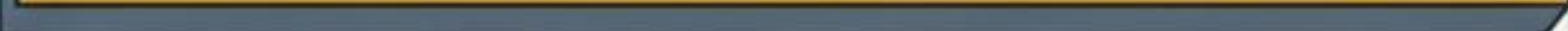
3. Mastering Sampling Techniques (Sourcing the components)



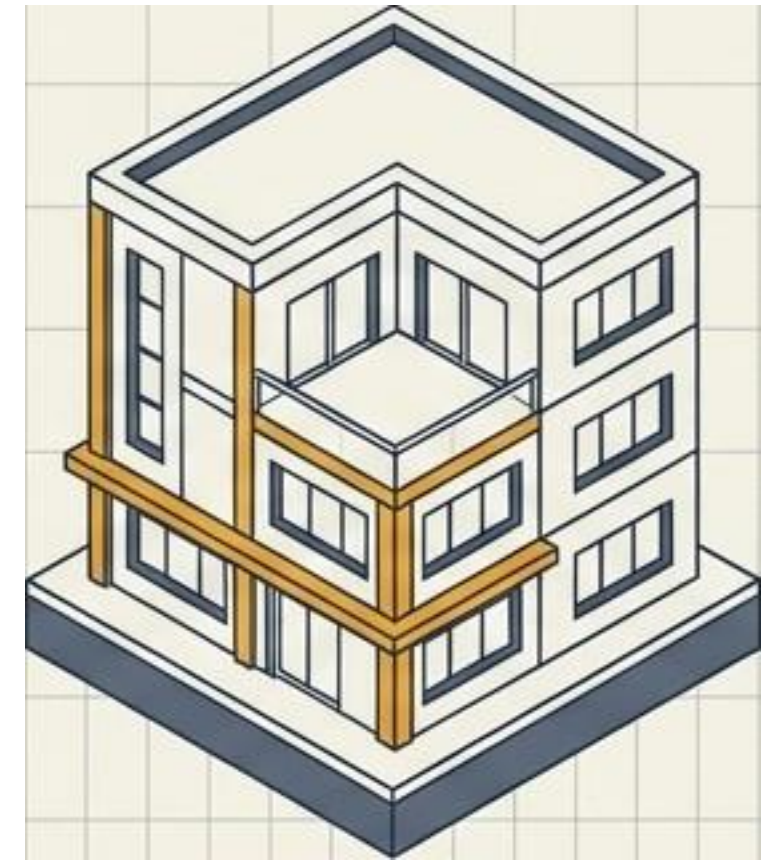
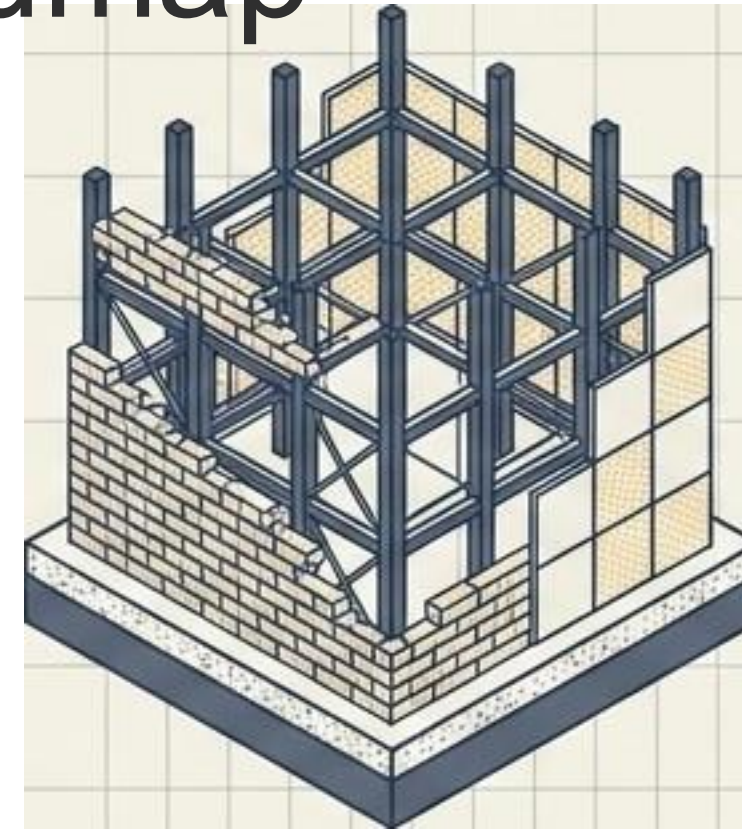
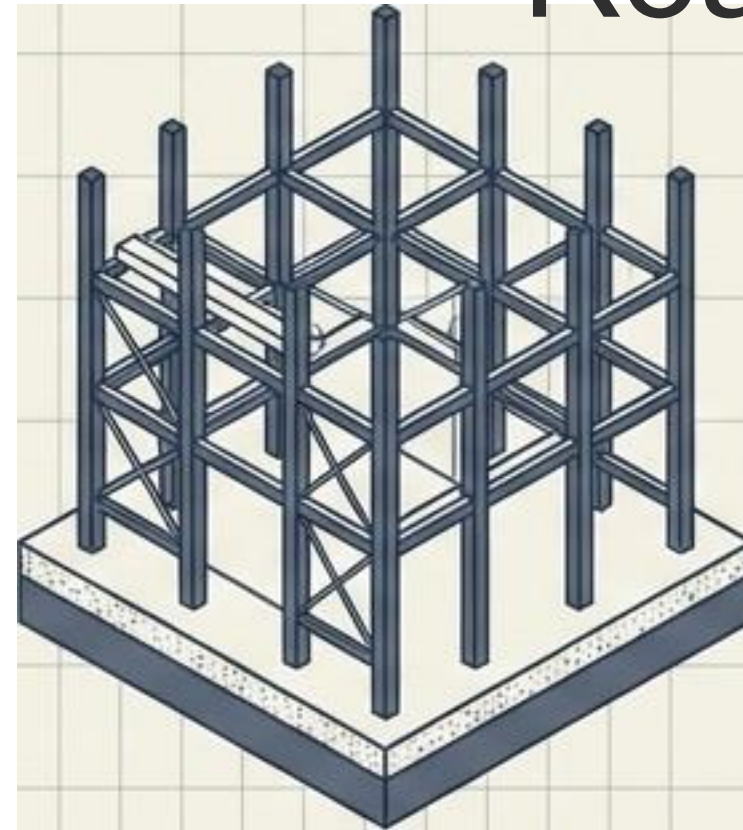
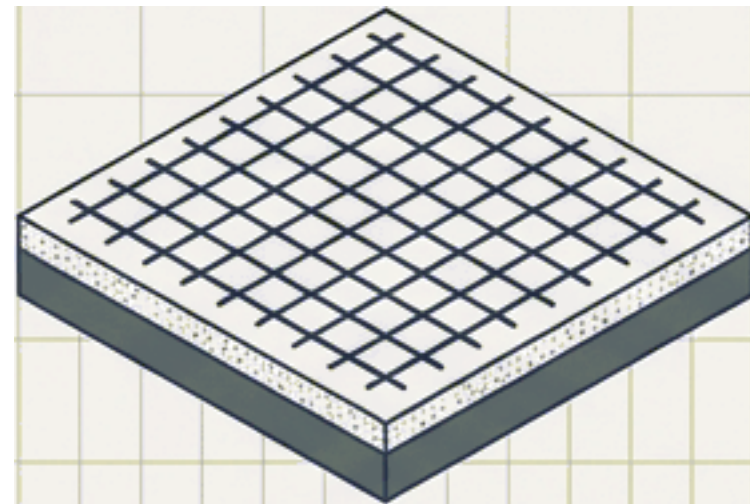
4. Conducting Literature Reviews (Reviewing historical zoning)



5. Developing the Research Proposal (The Final Blueprint)



Four Phases of Construction: The Syllabus Roadmap



Phase 1: The
Foundation (Unit 1)
Paradigms, Theory, and
Types of Research.

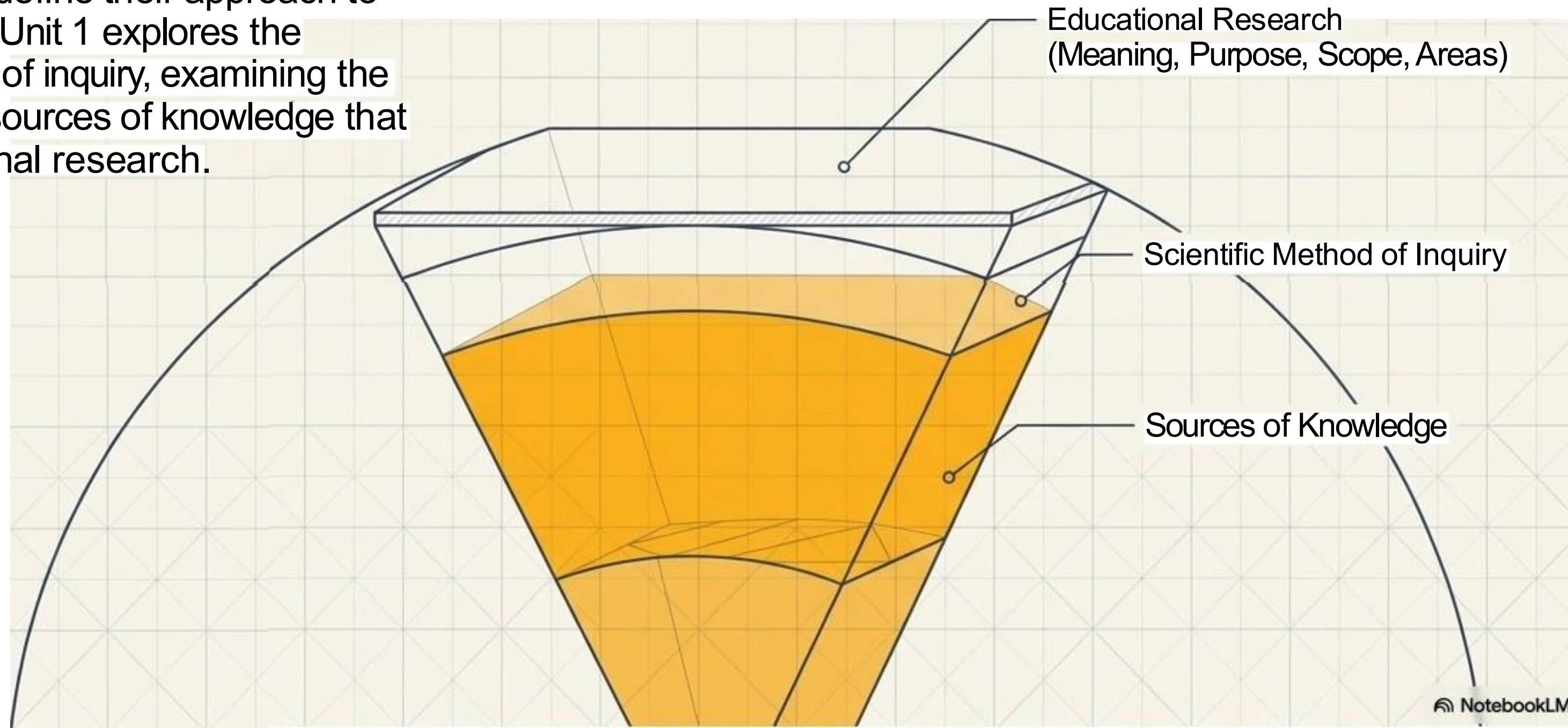
Phase 2: The
Scaffolding (Unit 2)
Review of Related Literature
and Problem Identification.

Phase 3: The
Materials (Unit 3)
Populations, Samples,
and Sampling Errors.

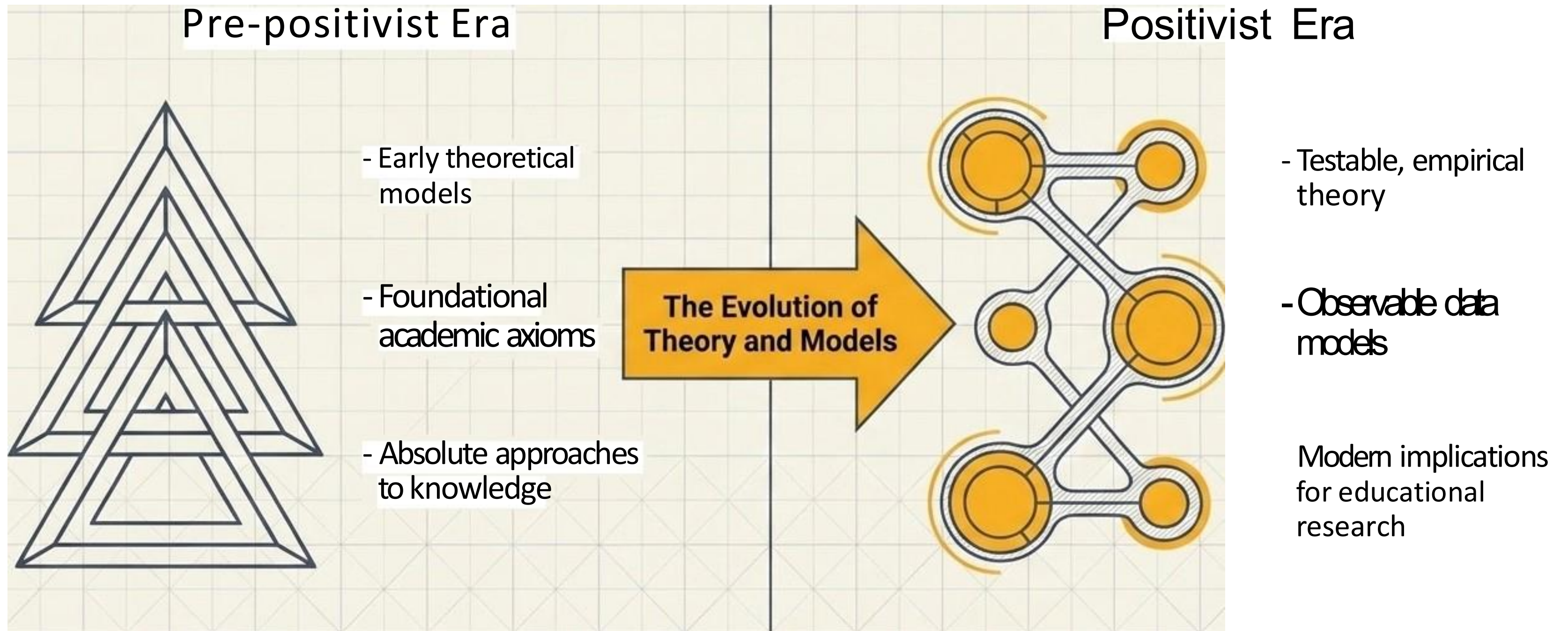
Phase 4: The
Assembly (Unit 4)
Objectives, Hypotheses,
and the Research Proposal.

Phase 1 Foundation: The Nature of Scientific Inquiry

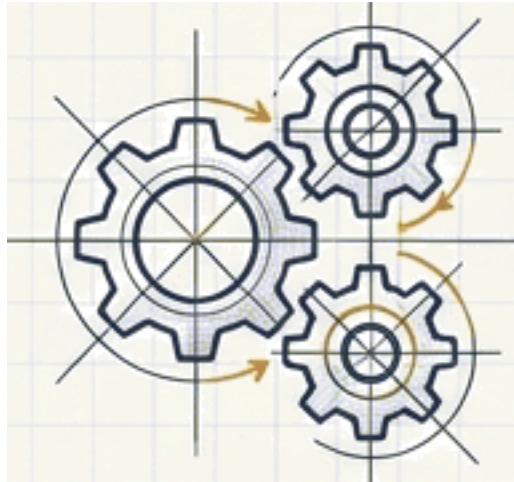
Before formulating a methodology, a researcher must define their approach to knowledge itself. Unit 1 explores the scientific method of inquiry, examining the very nature and sources of knowledge that validate educational research.



The Paradigm Shift: Evolving Lenses of Inquiry

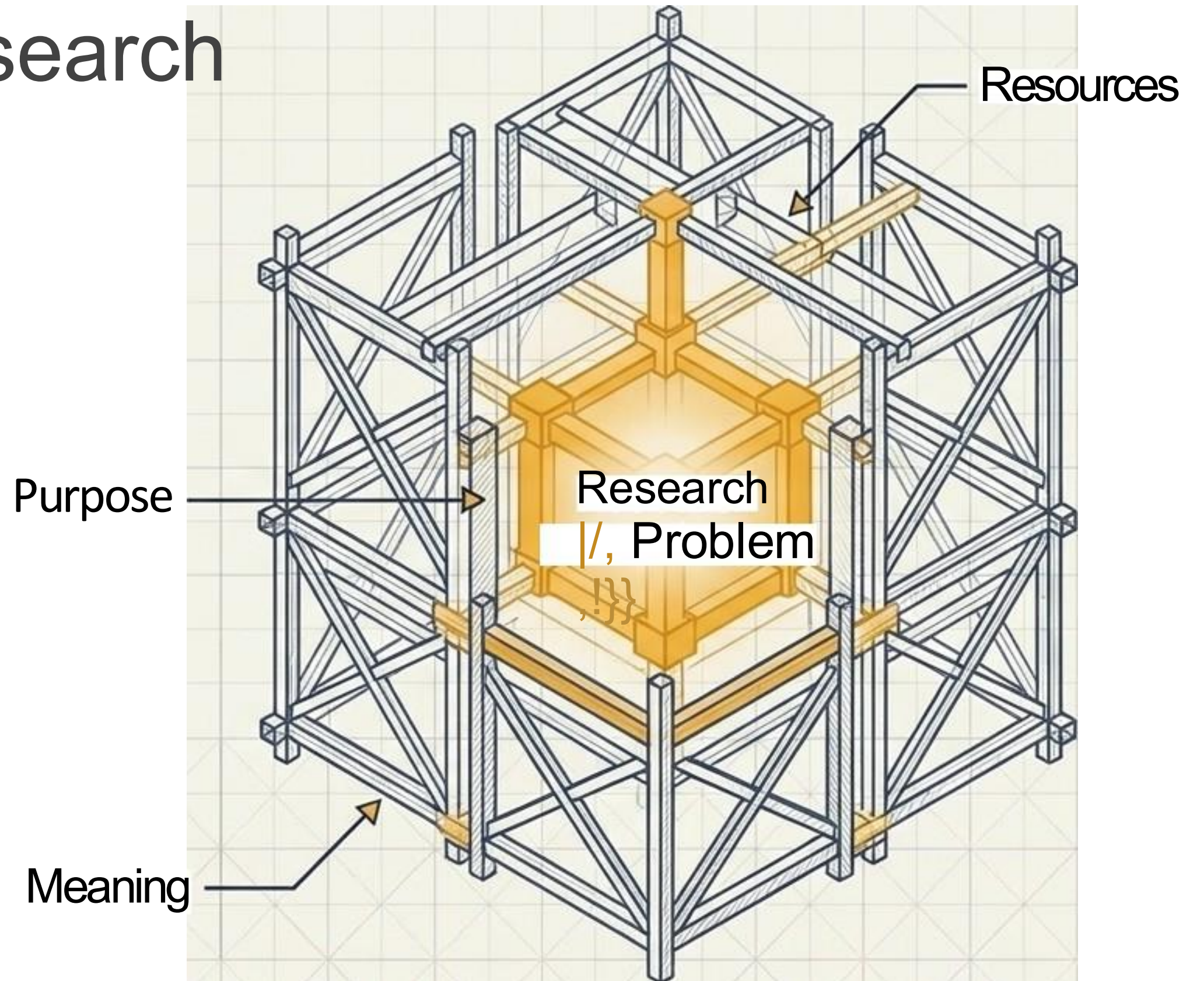


The Research Triad: Basic, Applied, and Action

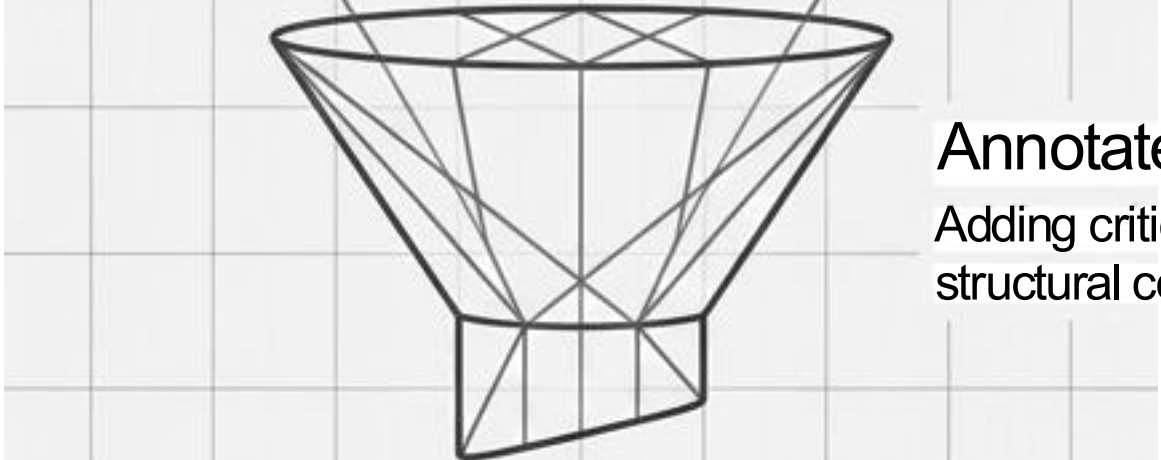
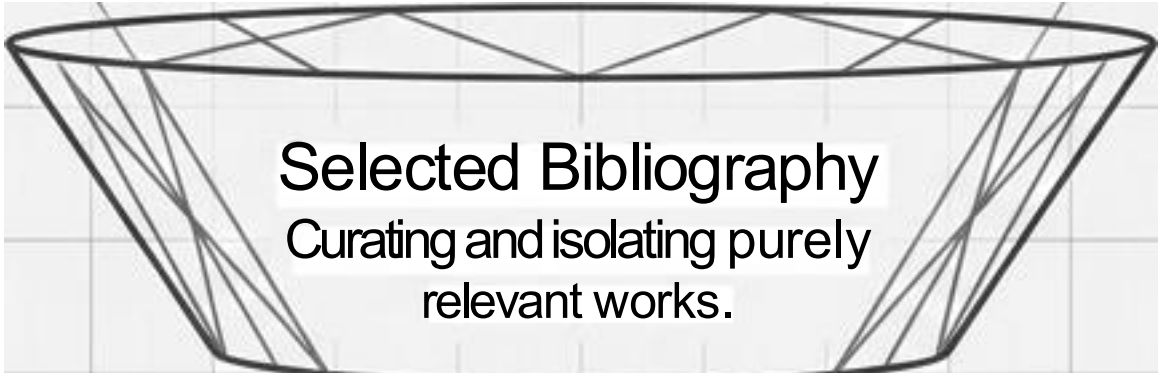
Basic Research	Applied Research	Action Research
		
Purpose: Theory generation and expansion. Scope: Universal and broad applicability. Characteristics: Driven by fundamental curiosity; seeks to establish foundational principles.	Purpose: Solving specific, practical problems. Scope: Targeted to specific fields or contexts. Characteristics: Focuses on testing and applying theories within real-world environments.	Purpose: Immediate, localized improvement. Scope: Hyper-local, often classroom specific. Characteristics: Conducted directly by practitioners; highly cyclical for immediate application.

Phase 2 Scaffolding: Contextualizing the Research

A research proposal cannot stand in a vacuum. Unit 2 provides the tools to conduct a comprehensive literature search using databases and the internet, ensuring your research is structurally supported by existing academic discourse.



The Literature Funnel: From Broad Data to Specific Problem



Identification of the Research Problem
The precise, actionable gap in the existing literature.

Quality Assurance and Academic Standards



Citation Architecture.
Mastering APA style for uncompromising structural consistency across all academic references.



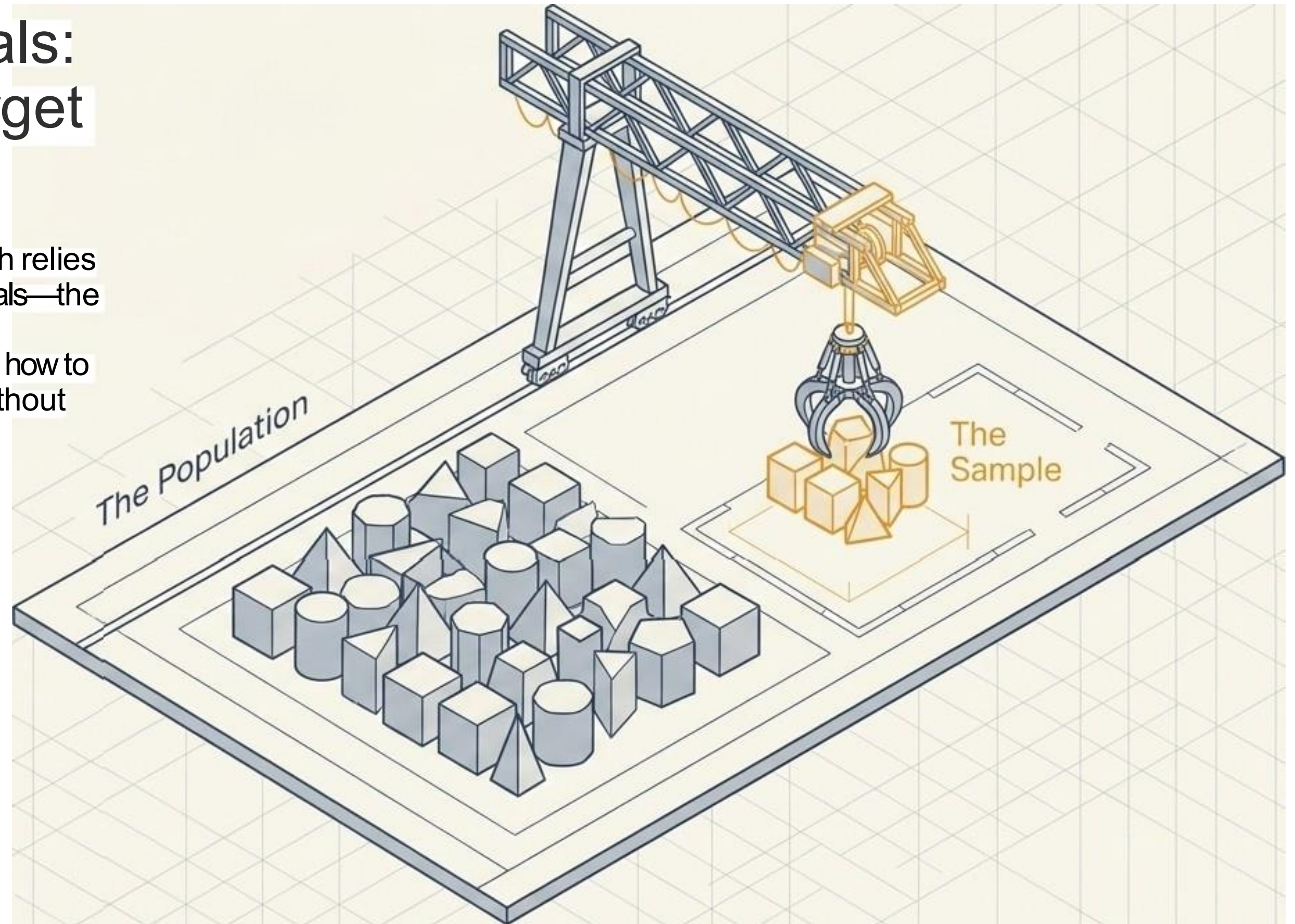
Source Integrity.
Rigorously evaluating the quality of internet resources to ensure foundational stability.



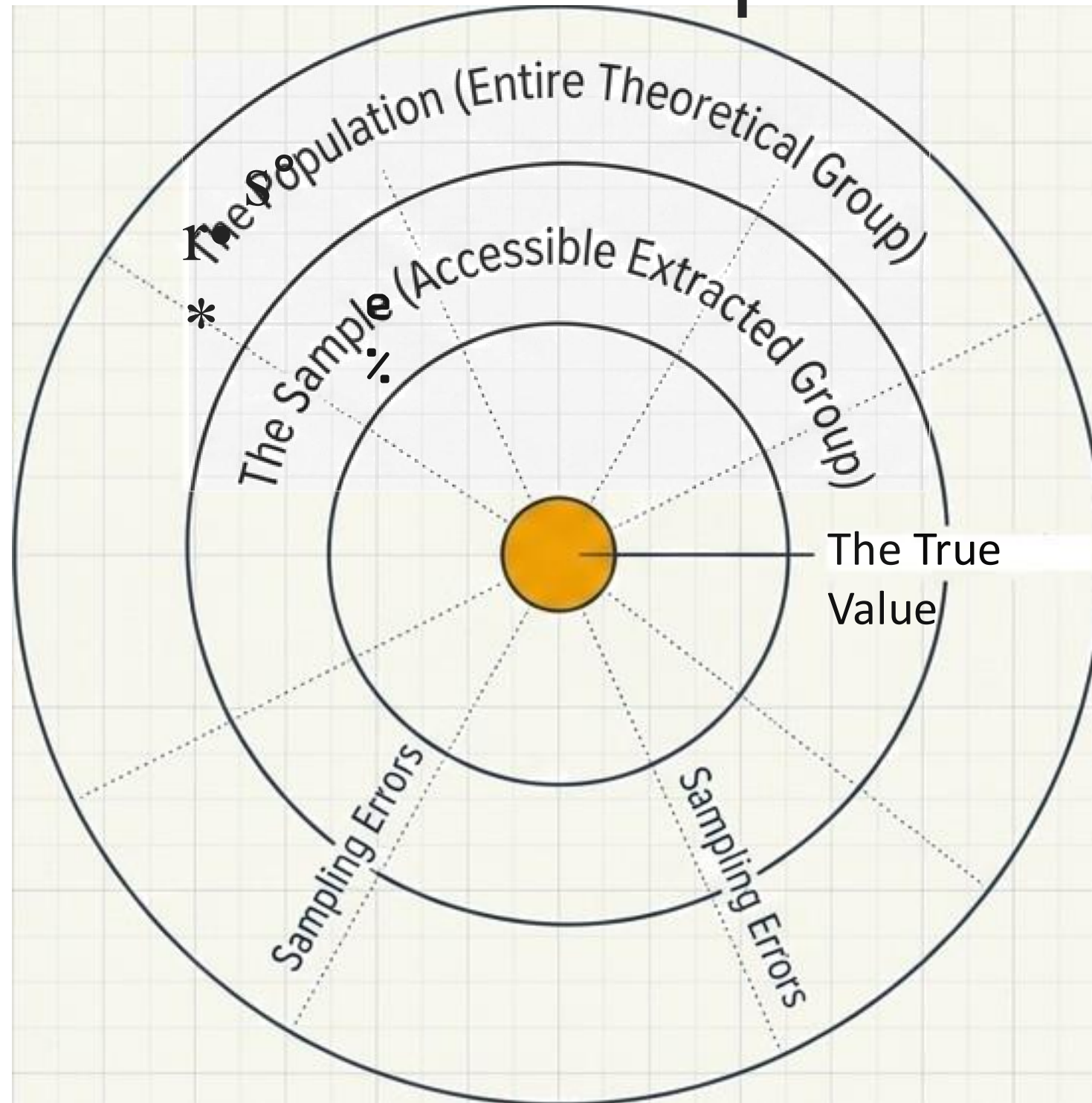
Tool Proficiency.
Utilizing specialized internet search tools and databases to locate verified academic materials.

Phase 3 Materials: Defining the Target Population

The validity of educational research relies entirely on the quality of its materials—the sample. Unit 3 deconstructs the characteristics of populations and how to extract representative samples without compromising structural integrity.



Precision and Variance: The Population Target



Ways to reduce sampling errors

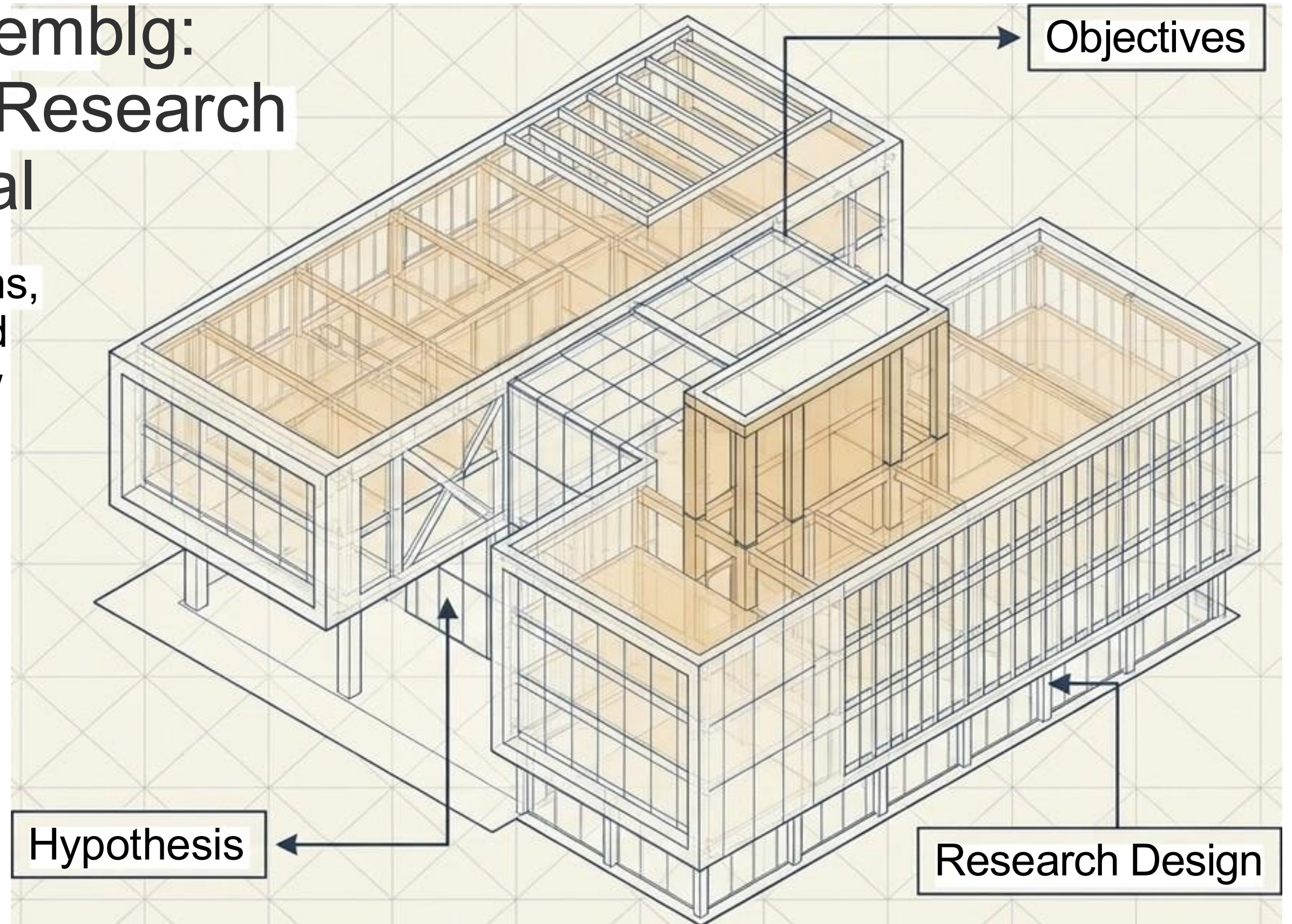
- - Increasing sample size
- - Improving randomization techniques

The Sampling Matrix: Selecting Your Extraction Method

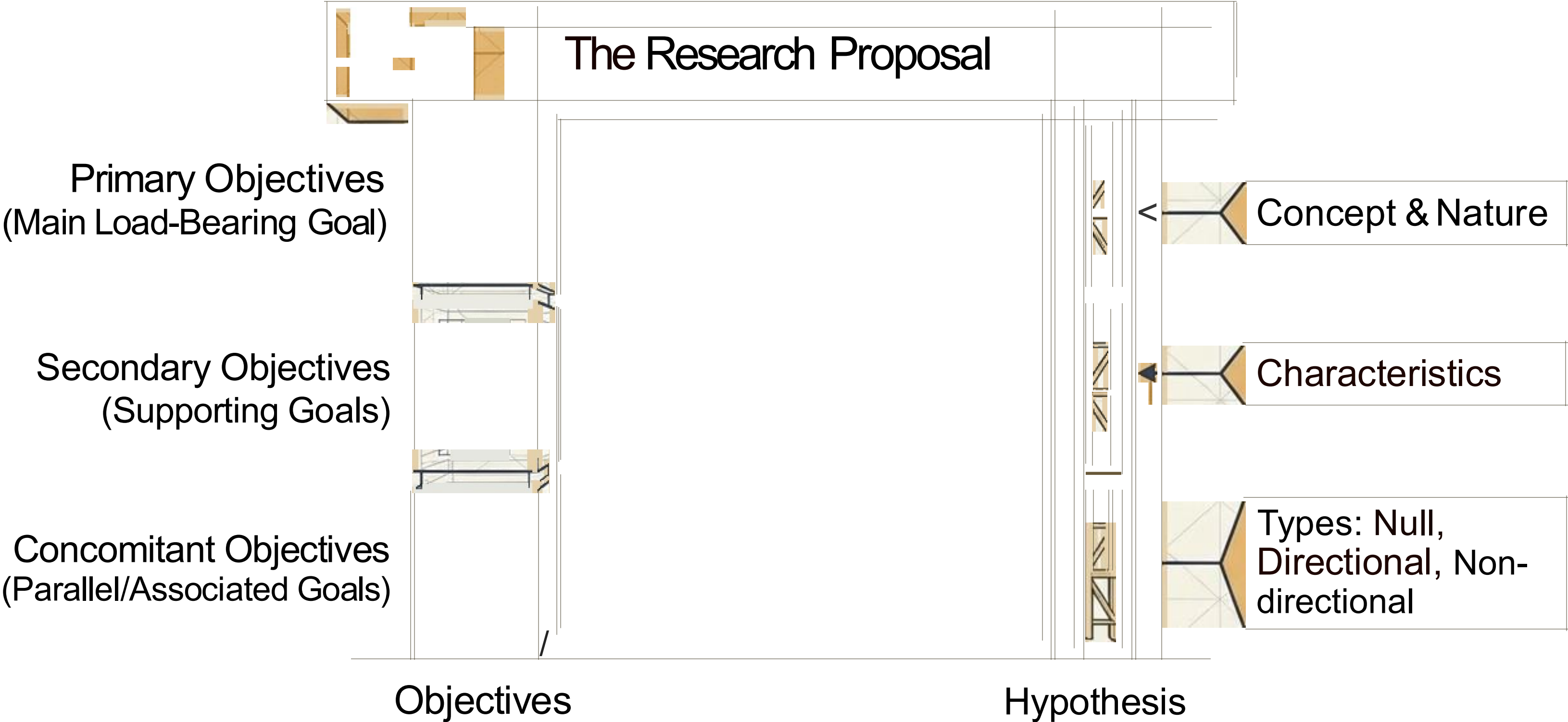
Probability Sampling	Non-Probability Sampling
	
- Random selection mechanisms. - Every unit has a known, non-zero chance of selection. - Ideal for quantitative, basic, and applied research. - Seeks broad generalization to the larger population.	- Subjective selection mechanisms. - Relies heavily on researcher judgment (e.g., purposive, convenience). - Ideal for qualitative and action research. - Requires specific localized contexts, not broad generalization.

Phase 4 Assemblg: Developing the Research Proposal

All foundational paradigms, literature scaffolding, and sample materials are now assembled into a singular, cohesive architectural document: The Research Proposal.

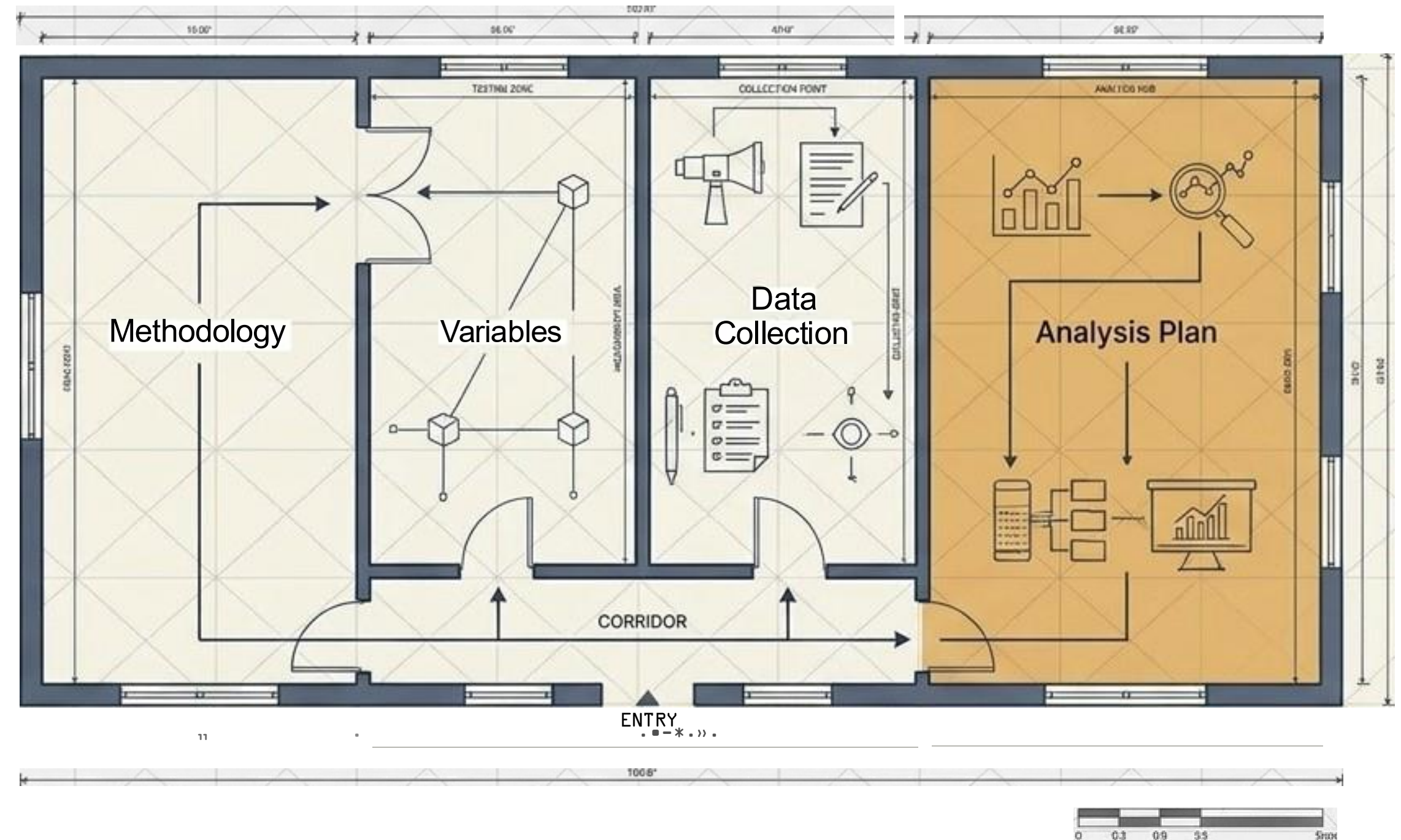


Engineering the Core: Objectives and Hypotheses

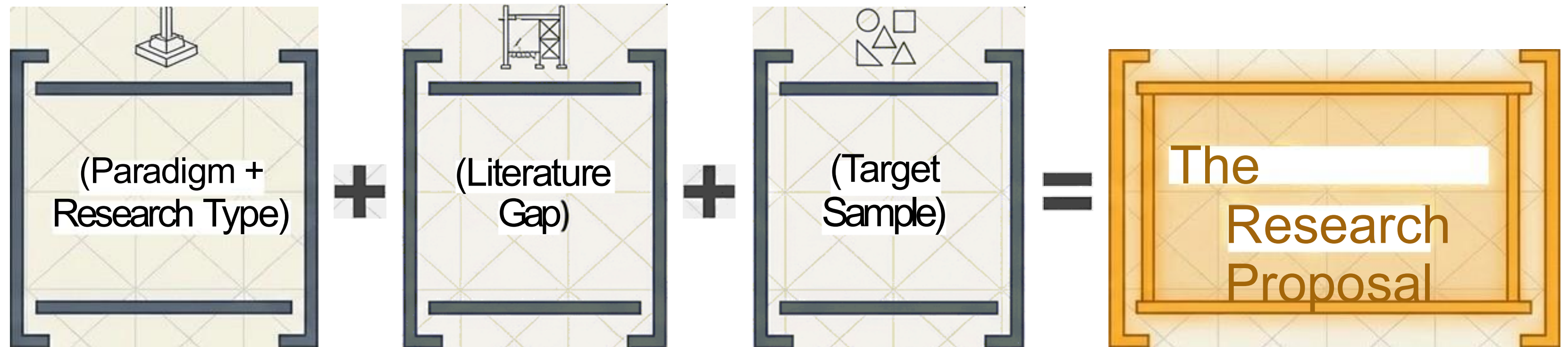


The Final Blueprint: The Research Design

The Research Design is the definitive schematic. It dictates exactly how the hypothesis will be tested, how the sample will be measured, and how the objectives will be met, ensuring the entire structure is logically sound and empirically valid.



The Assembly Diagram: Synthesizing the MED104 Framework

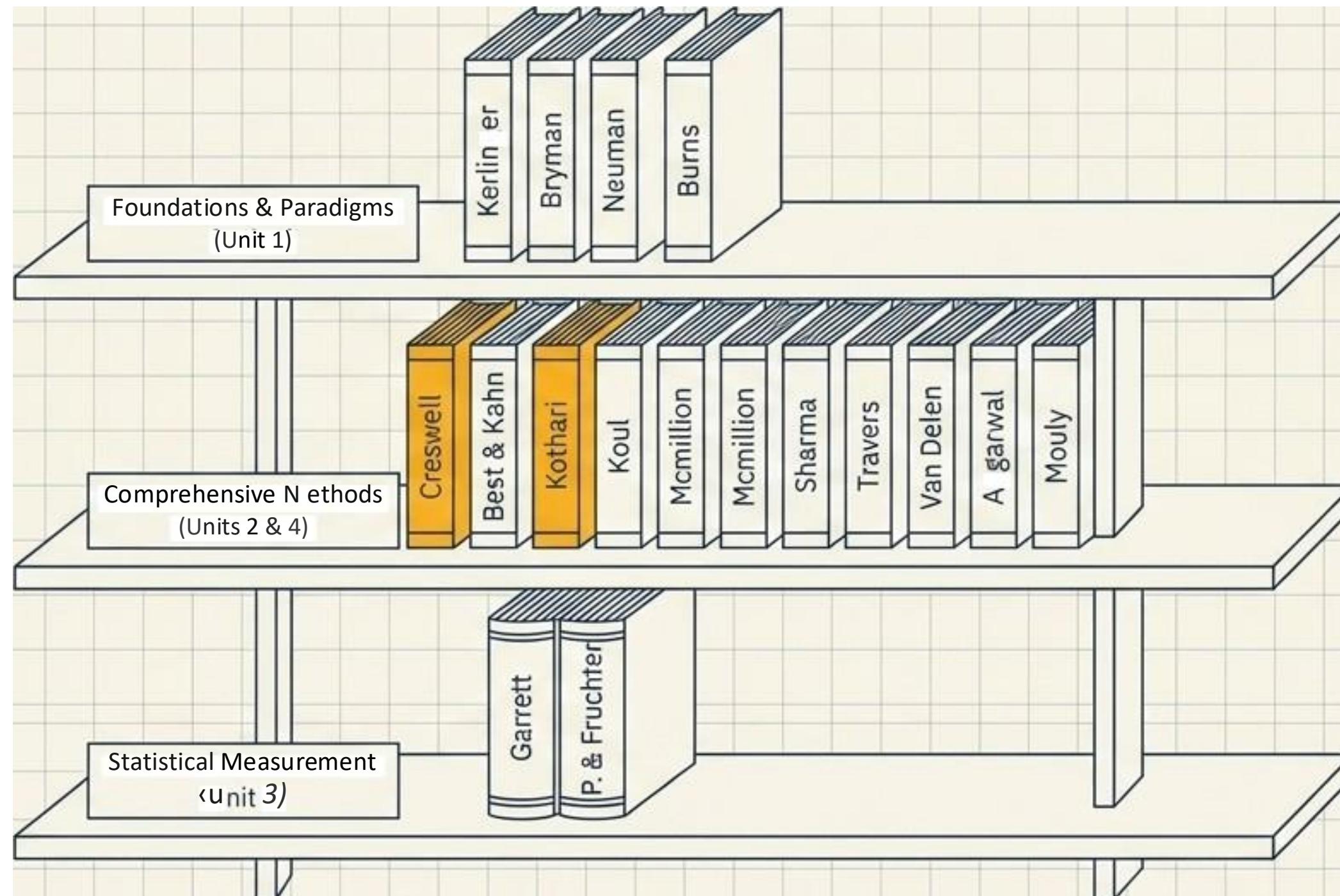


A proposal cannot exist without alignment. A positivist paradigm requires probability sampling. An action research type demands highly specific literature contexts. Every choice dictates the final design.

Course Evaluation Blueprint



The Scholar's Bookshelf: Curated Resource Materials



From Consumer to Architect

You now possess the blueprint, MED104 is not just about understanding how past knowledge was built; it is about equipping you to construct the future of educational inquiry with precision, rigor, and purpose.

